Forecasted BG Sequence

Glucose $G^{(k+1:k+H)}$

Final Conv1D

Decoder Block N
2×Conv1D
(filter size f, number of channels: $\tilde{c}_N$)

Upsample

....

Upsample

Decoder Block 1
2×Conv1D
(filter size f, number of channels: $\tilde{c}_1$)

Decoder

Encoded Latent Sequence

Encoder Block N
2 × Conv1D
(filter size f, dilation $\delta_N = b^{N-1}$, number of channels: c_N)

AvgPool

....

AvgPool

Encoder Block 1
2×Conv1D
(filter size f, dilation $\delta_1 = b^0$, number of channels: c_1)

Encoder

Glucose $G^{(k-d_G:k)}$

Meal $M^{(k-d_M:k)}$

Insulin $I^{(k-d_I:k)}$

Multichannel Input Sequence

Temporal Residual Block

+

ReLU

Causal Conv1D (f, δ_j, c_j)

1×1 Conv1D

ReLU

Causal Conv1D (f, δ_j, c_j)

Bayesian Causal Conv1D

c_j

dot product

w_f ... w_2 w_1

c_j

c_{j-1}

Input Feature Tensor